

Dhruv Patel

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Education

- **Stony Brook University** Aug 2025 – May 2027
Master of Science in Computer Science, GPA: 3.61/4.00
Stony Brook, NY
- **Dhirubhai Ambani University** Nov 2020 – May 2024
B.Tech. (Honours) in ICT, Minor in Computational Science, GPA: 3.45/4.00
Gandhinagar, India

Projects

- **Aaron Technologies: Website for B2B Industrial Sourcing Platform** *Next.js, React, TypeScript, Firestore, Resend, HubSpot*
 - Built production features for a B2B industrial sourcing platform, architecting a **serverless backend** using Firestore, Resend, and HubSpot CRM for data persistence, transactional email, and automated lead synchronization.
 - Developed an interactive **landed-cost calculator** modeling freight, duty, and handling, alongside **15+ responsive pages** spanning product, catalog, industry, and RFQ workflows using Next.js, React, and TypeScript.
 - Improved production readiness through site-wide **WCAG 2.1 AA accessibility**, security, SEO, and performance improvements, including JSON-LD structured data and deployment hardening.
- **ConsensusPrompt: Multi-Agent Middleware for Prompt Optimization** *LangGraph, FastAPI, Next.js, OpenRouter*
 - Engineered a multi-agent middleware applying anonymous peer review to prompt engineering, coordinating **rewriter, reviewer, and chairman agents** through a stateful LangGraph orchestration workflow.
 - Validated the system in a **13-user study**, achieving **4.77/5 mean trust** with a 0% override rate; an ablation study showed a **25.5% relative gain** in consensus strength from a 5-model reviewer configuration.
- **OnboardOps: AI-Powered Repository Onboarding Platform** *FastAPI, Next.js, MCP, GitHub API*
 - Built an AI-powered developer onboarding platform with a **custom MCP server** and FastAPI backend to automate understanding and exploration of unfamiliar, large software systems.
 - Engineered real-time WebSocket workflows and repository retrieval pipelines using GitPython and GitHub APIs to identify dependency graphs, entry points, code hotspots, and project conventions.
- **ARGUS: Local AI Site Compliance Platform** *Python, FastAPI, CUDA, PyTorch, React*
 - Architected a fully on-device, privacy-preserving **multimodal AI platform** orchestrating five models to ground construction-site video findings in OSHA and NYC DOB regulations through retrieval-augmented search.
 - Integrated **7 live NYC Open Data datasets** for compliance enrichment and optimized local **CUDA inference** through detector reuse, VLM batching, SDPA attention, and startup prewarming.
- **Freight Room: Multi-Agent Supply Chain Response Platform** *Winner – Band of Agents Hackathon 2026*
 - Built a distributed platform that detects vessel delays from a **live AIS feed** and dynamically recruits **7 specialized agents across 3 frameworks** (LangGraph, PydanticAI, CrewAI) to coordinate logistics incident response.
 - Designed an auditable human-in-the-loop workflow using quorum voting, adversarial dissent review, and demurrage estimation grounded in published carrier tariffs to produce traceable decision dossiers.
- **Raft Consensus & Multi-Paxos Sequencer** *C++, Raft, Multi-Paxos, MAKO, Masstree*
 - Implemented Raft consensus with leader election, replicated logs, and fault-tolerant recovery for a distributed key-value system, validated under simulated node failures and network partitions.
 - Extended the system with a deterministic-ordering transaction engine using a global sequencer and Multi-Paxos to guarantee strictly serializable execution, achieving **200+ TPS across 3 replicas** and **390+ TPS on a single-replica baseline** using the TPC-C benchmark on MAKO.

Research Experience

- **Research Project** Jan 2024 – Jul 2024
Integrating DL with 2D-3V Particle-in-Cell Simulations of Low Temperature Plasmas (APS GEC 2025) *DA-IICT, India*
 - Accelerated device-scale **Particle-in-Cell (PIC-MCC) plasma simulations** by replacing charge-deposition, Poisson-solver, and electric-field stages with a **U-Net surrogate model**, achieving a **45% execution-time reduction** on 128x128 grids with 2.6×10^5 particles.
 - Enforced physical fidelity using **Physics-Informed Neural Networks**, maintaining mean percentage error **under 5%** against baseline PIC-MCC across 500-iteration ExB plasma test cases; implemented in Python with C extensions on Unix/Linux.
- **Summer Research Intern** May 2023 – Aug 2023
Software Testing and Test Case Prioritization *DA-IICT, India*
 - Implemented and benchmarked four test-case prioritization techniques across **30 open-source Java projects** ranging up to 87K SLOC and 5,000+ test cases, using runtime and mutation-based APFD evaluation.
 - Profiled execution bottlenecks and automated evaluation pipelines in Python, **reducing evaluation pipeline runtime by 30%**.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, MATLAB, R

AI / ML: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, LangChain, LangGraph, HuggingFace, FAISS, CUDA, OpenRouter, MCP

Backend / Web: Next.js, React, FastAPI, Flask, Node.js, REST APIs, WebSockets, Server-Sent Events

Systems: Distributed Systems, Concurrency, Multithreading, Socket Programming, System Design, Object-Oriented Design, Performance Optimization, Unix/Linux

Cloud / DevOps: Docker, Kubernetes, AWS, Azure, Google Cloud Platform, Git, GitHub, CI/CD, pytest, JUnit, Vercel

Databases: PostgreSQL, Firestore, MongoDB, Redis, MySQL, Supabase

Leadership

- **Deputy Convenor** Apr 2021 – Apr 2023
Annual Techno-Cultural Festival Committee *DA-IICT, India*
 - Led the technical team for one of Gujarat's largest techno-cultural festivals, delivering infrastructure and event execution for **25,000+ attendees**.